Code No: 117HN

R13

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations April/May - 2018
SOFT COMPUTING
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a) What is state space? [2]
b) How to define a problem for AI? [3]
c) State the delta rule for basic ANN. [2]
d) Give the architecture of a simple perceptron network. [3]
e) What is the fundamental operating principle of adaptive resonance theory network? [2]
f) What is the significance of Cauchy machine? [3]
g) What is meant by characteristic function? [2]
h) Compare and contrast classical logic with fuzzy logic. [3]
i) Define the belief measure. [2]
j) Why is approximate reasoning important in fuzzy logic? [3]

PART-B

(50 Marks)

2.a) Compare depth first search with breadth first search.
b) What is meant by a heuristic search? [5+5]

OR

3. Explain hill climbing algorithm and simulated annealing with suitable examples. [10]

4.a) What is the impact of weight in an artificial neural network?
b) What are the basic models of an artificial networks? [5+5]

OR

5.a) What factors influence the convergence of back-propagation network?
b) Compare BAM and Hopfield networks. [5+5]

6.a) What are the important features of Kohonen self organizing maps?
b) State the applications of Kohonen self-organizing maps. [5+5]

OR

7. Describe the architecture of Boltzmann machine and give the testing algorithm for it. [10]

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8.a) Obtain fuzzy relation T as a composition between the following fuzzy relations

R	y1	y2
x1	0.6	0.3
x2	0.2	0.9

S	y1	y2
x1	0.6	0.3
x2	0.2	0.9

b) Define fuzzy equivalence relation. [5+5]

OR

9.a) Describe how neural network is used to obtain fuzzy membership functions.

b) Which defuzzification technique is preferred? Why? [5+5]

10. What are the five functional blocks of fuzzy inference system? Explain its construction and working methodology of fuzzy inference system. [10]

OR

11.a) Explore application of genetic algorithms to travelling sales person problem.

b) How to compute Jaccard's score based on links and indexing? [5+5]

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